Final Project; Coding Supplement Guidelines

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This document summarizes the final project guidelines that we talked about in class.

IMPORTANT:
The word limit for this project is 1500 new words and is strictly enforced. Less is better, more is not. You have 1500 word count for P1 and 1500 extra for P2, and 1500 for the final project; summing to 4500. Exceeding the limit may result in a penalty of up to 20%. To avoid this, ensure your writing is concise and do not generate irrelevant text using ChatGPT (plus academic integrity issues for fully AI-generated projects). Heavily ChatGPT generated text that are specifically not backed by the project results may result in up to 100% penalty.
AI policy:
Students may use AI tools (e.g., ChatGPT, Copilot) for brainstorming or editing assistance. However, all analysis, interpretation, and writing must reflect the student's own understanding. Students remain responsible for verifying the accuracy of any information produced with AI assistance. In particular, if an AI tool suggests an academic reference, you must confirm that the paper exists and that it supports the claim you attribute to it. Citing non-existent papers or misrepresenting the contribution of a cited paper will be treated as an academic integrity violation.

Mandatory Section Titles in Your Final Project

Please note, the section titles in the "Final Paper" will be different.

You can and should add more subsections and sections when needed, but all projects must have the following "main" section titles in the same order. We will keep updating this list for future projects.

Your Final project should include all your previous projects. You will have three main sections for your project (and some subsections that you title yourself): "Project One," "Project Two," and "Final Project." You can add titles by adding "#" sign before your title in your markdown cell; check this link for more details on titling: https://www.datacamp.com/community/tutorials/markdown-in-jupyter-notebook

1. Project One (Use # in a Markdown Cell to set this as a main section)

1.1 **Introduction** (use ## in Markdown cells to set the following as the subsections)

1.2 Data Cleaning/Loading

1.2.1 you can have a subsection for new dataset(s) that you merge to explain them separately

1.3 Summary Statistics Tables

1.4 Plots, Histograms, Figures

2. Project Two (Use # in a Markdown Cell to set this as a main section)

    2.1 The Message

    2.2 Maps and Interpretations

    2.3 Regressions

3. Final Project (Use # in a Markdown Cell to set this as a main section)

    3.1 Potential Data to Scrape

    3.2 Potential Challenges

    3.3 Scraping Data from a Website

    3.4 Visualizing the Scraped Dataset

    3.5 Regression Tree

    3.6 Random Forest

    3.7 OLS vs ML

    3.8 **Conclusion** (note that the entire document has one conclusion, not two or three. Same for the introduction)

Formatting

Projects lose at least **15 points** if submitted project does not meet ANY of the following requirements:

1. Submitted projects must be in PDF. Any other format, including a .ipynb file, is not accepted.

2. The projects should have clear "section titles." Besides the subsections you choose to have, you should have three main sections; Project One, Project Two, and Final Project.

3. Clear title for your entire project.

4. You should write your project in Jupyter Notebook (Python) and submit it in pdf format. If you have problems converting your notebook to pdf, first download it as HTML, then print/save the HTML version in pdf.

5. This is an individual project. However, you are encouraged to check the projects on the Kaggle.com website that use similar data. We have provided some "useful links" on the data list on Quercus. You can use these sources, but the coding and explanations **must be yours**. Do not copy and paste the same chunk of code in your project.

Please note that if you include a graph or a table, you should **explain what you learn from it.** Do not add an output without any explanation. All projects should have an introduction and a conclusion. Suppose you want to send your project to a company or school that you are applying to. The final product should be a clean and comprehensive report.

- Do not include unnecessary chunks of codes or outputs and errors.
- Any graph, summary, or output should have an explanation following. Why do you include it in your report, and what do you understand from it?

Part One: Introduction and Conclusion

(5 points) Introduction: As always, you should update your intro and conclusion according to your last results. Your introduction should be similar to an academic paper and of similar quality. Your paper should be publishable.
In your introduction, cite the relevant literature (a minimum of 10 papers), how your paper adds to the existing literature, and how it differs. If you cite a paper, you should explain why that paper is relevant and how citing that paper is contributing to your discussion. Simply listing papers and citing random sources will not receive marks here. The expected length for your introduction is a minimum of one and a half pages. Papers with few citations or no literature review lose a minimum of 5 to 10 marks.
(5 points) Conclusion and Future Work: As always, you should update your conclusion according to your last findings. Your conclusion should be similar to an academic paper's conclusion and of equal quality. You should include your findings and summarize your paper (at least one page). Once more, your paper should be publishable.

Future work: explain how you want to improve your paper in the future. Think about whether you need to add more data or try different types of regressions. Talk about the limitations of your work and the next steps to improve these limitations. This part should be backed by evidence and be in scientific writing. Vague and general writing is not acceptable. A chat GPT generated text or text that does not show thoughtful evaluation of the feasibility of the ideas will not receive any marks.

Part Two: Previous Projects

(30 points) Incorporate the comments that you have received in your previous projects. Finish the parts that you missed in your previous projects, even if you did not lose marks for them. This is the first part of your final project, which will be graded again. We will check the first section of your projects (your updated previous project) and grade it again. If you have not received any comments, try to improve your previous report by doing a better job on the introduction or literature review, fine-tuning your visualizations, or adding more meaningful analysis.

You should significantly improve the quality of your previous projects compared to your initial submissions. Simply printing the previous projects will not be acceptable and such work will not receive the points for this part.

Final Project Details (New)

You are almost there. We built upon the past projects towards a complete academic paper on a hot topic using real-world data and cool Python techniques, just excellent! Here's what we are going to do for your final project.

IMPORTANT: **Think hard and complete ALL of the parts below (and answer all questions). Show your code and results, and provide economic explanations for your results for all parts and connect them to your research question and main message. We will evaluate your work's quality and accuracy; simply answering all questions does not guarantee a full mark.**

Part Three: Web Scraping

(30 points)

1. **(5 points)** Think about the main message/question of your paper. Think about what information you can add to your dataset that is not already in your dataset and requires web scraping. Know that there are many other available datasets you can download, and you do not need to do web scraping to acquire them. So, focus on the data you can have if you generate it with web scraping.

   If you can find the answer to this question and work on it in the future, you will probably be the first one working on that topic! It can lead to an original paper. So, describe the website that can be an excellent addition to your dataset and clearly explain how it can improve your results. It may be HTML-based or API-based scraping (searching for available scraping APIs is an excellent place to look for new ideas and accessible tools for web scraping).

   Please note using APIs for datasets that can be downloaded as excel files such as Census, BLS, FRED, and alike does not count and is not what we are looking for. If you are using an API, it should be from a data that is not easily downloadable such as Twitter.

   Clearly explain (1) why you think this data enhances your paper, (2) the address of the website where you scrape it from, (3) what data you are scraping from which part of the website (4) how you can or cannot merge the data with your results, and how you are going to use this new information in answering your research question.

2. **(5 points)** What kind of challenges you are/may be facing when scraping this source. Do you need to run your program over time (daily, monthly, annual) to generate the data? Can you scrape the data source you have found for this project? If not, can you scrape parts of it and visualize?

3. **(10 points)** Scrape your data, show your code. I will add extra points for brilliant ideas (they will be based on the originality and quality, extra marks not guaranteed just by completion).

   The website you are choosing should not be easy to copy and paste; it should have a time component that requires frequent scraping or some other reason why using web scraping is necessary.

   - Scrape the data from your chosen website. Show your code.
   - You should clearly explain each chunk of your code in this part. Maybe divide your scraping code into three (or more) parts as we did in the lab and add explanations for someone new to web scraping after each part. Simple one line API web scraping does not get full marks for this part. We are practicing to work with HTML and deal with potential challenges. If you use an API, it should not be a data download API. Check with me if you are not sure about your source.

4. **(10 points)** visualize the new scraped data and explain how it is related to your project (use maps, trends, histograms, etc.). One visualization is the minimum requirement and gets the average credit for this section. For full credit, create more than one visualization type or add a well-structured table followed by economic interpretations. Always tie your interpretation back to the research question.

   IMPORTANT: Do not merge if you are not sure that you can keep **ALL** of your observations in your **MAIN** dataset. Ensure that you don't discard any observations during the merging process if you decide to merge. The original datasets I provided you contain thousands and occasionally millions of observations. If your analysis yields less than 1000 observations, depending on the nature of your project, your final project and future regressions may not receive any credit (the only exception is if your original provided data had less than 1000 observations). I provided extensive datasets to enable you to produce important research papers and powerful regressions. If you drop all of your observations and end up with (for example) only around 30 data points, your final project's regressions will not be evaluated and will receive no credit. *Please talk to me during office hours or after class if you have any questions.*

Part Four: Machine Learning

Optional: If that makes sense, you can add a regression table with new/updated regressions here. There will not be extra points for this; however, it may help your writing. Also, if you have new causal regressions (DID, IV,...), you can add a new section called the "causal model" and include the regressions plus your interpretation. And then you can move on to your ML section.

(30 points)

1. This part is harder and may take more time than expected. So, make sure you have done a good job in the previous parts first. I do not want you to spend too much time on this part before having a good introduction and OLS part.

2. Re-write the objective function for a regression tree with your variables and clearly explain the objective function in your own words.

3. Explain the regularization parameters and talk about how changing them can affect your model and results.

4. Run the regression tree using the Xs you can justify including in the model. They can be coming from your preferred regression.

5. Run another regression tree with all of your X variables. Are the chosen variables and splits different in this tree compared to the preferred specification tree? What can you learn? This is a kitchen sink (all of your X variables except zipcodes, etc)

6. Output your tree and explain your results in economic terms. I provide example explanations and discussions in class.

7. Talk about your error of prediction.

8. Run a Random Forest Model and interpret the importance matrix. Is the results of the important matrix teaching you anything new/surprizing/different? What insights are you getting from this matrix?

9. **(5 out of 30 points for this part)** Compare your results from running a regression with your results from running a regression tree. What extra information have you extracted using the regression tree that was not possible to extract with a linear regression? Your explanations should tackle both the econometrics of the OLS and regression trees and the economic intuition behind both models.

Upload your Jupyter Notebook (your code and explanation) in pdf format on Crowdmark for marking. Also, please push your final Jupyter to Git.

Well, you made it after all! :)